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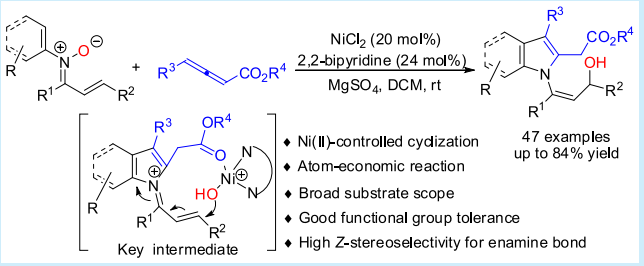
Nickel(II)-Catalyzed [3 + 2] Cycloaddition of Nitrones and Allenoates to Access *N*-Vinylindoles and *N*-Vinylpyrroles

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ABSTRACT: A variety of *N*-vinylindoles and *N*-vinylpyrroles were prepared in moderate to good yields through the nickel(II)-catalyzed [3 + 2] cycloaddition of α,β-unsaturated nitrones with allenoates under mild reaction conditions. A rational mechanism for the formation of *N*-vinylindoles was proposed based on the ^{18}O-labeled experiments and key intermediates detected by high-resolution mass spectrometry trace experiments. The present method highlights a nickel(II)-controlled cyclization, atom-economical reaction, broad substrate scope, good functional group tolerance, and high Z-stereoselectivity for the enamine bond.

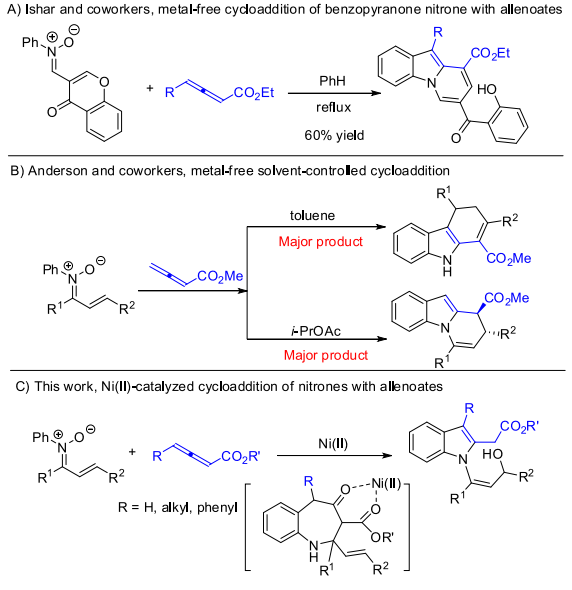


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N-Vinylindole derivatives not only are extensively found as core structures in various natural products and pharmaceuticals, such as apovincamine and apotacamine,[1] but also widely serve as valuable intermediates in various reactions or monomers for photoactive and polymeric materials.[2] Many strategies for the synthesis of *N*-vinylindoles have been developed in the past decades, including the stereospecific Pd-catalyzed *N*-vinylation of indoles with vinyl triflates or vinyl halides,[3] Pd-catalyzed oxidative cross-coupling of indoles and alkenes,[4] condensation of indoles with ketones,[5] intermolecular addition of indoles to alkynes,[6] Pd-catalyzed coupling of N−H indoles with *N*-tosylhydrazones,[7] and cyclization of *o*-alkynylaniline derivatives.[8] However, these reactions have usually suffered low efficiency and required a high temperature, expensive transition-metal catalysts, or strict reaction conditions. Hence, the development of new strategies toward the synthesis of *N*-vinylindoles from easily available starting materials and inexpensive metal catalysts is valuable.

Nitrones, serving as important 1,3-dipoles, have been one of the most attractive building blocks for building indole scaffolds or other heterocycles.[9] In the last two decades, α,β-unsaturated nitrones have attracted great attention in the preparation of complex target molecules owing to the rich transformations of the α,β-unsaturated bonds in nitrones during their reaction with other dipolarphiles.[10] In particular, the cycloaddition of α,β-unsaturated nitrones with allenoates was investigated.[11] For example, in 1999, Ishar and coworkers developed the reaction of benzopyranone nitrone and allenoate to access benzoindolizines through a metal-free cascade of a [3 + 2] cycloaddition/[3,3]-rearrangement/retro-Mannich reaction/Diels−Alder reaction (Scheme 1A).[12] In 2014, Anderson and coworkers pioneered a metal-free solvent-controlled strategy to prepare dihydrocarbazoles and dihydropyridoindoles, respectively, from *N*-aryl-α,β-unsaturated nitrones and unsubstituted

Scheme 1. [3 + 2] Cycloaddition of Nitrones with Allenoates



allenoates (Scheme 1B).[13] Later, an asymmetric synthesis of dihydropyridoindoles was achieved through a bifunctional thiourea catalyst.[14] A facile method for the synthesis of 1,4-

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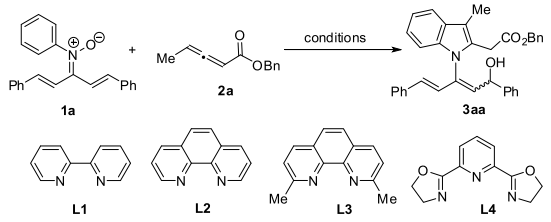
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enaminoketones was reported by the [3,3]-rearrangement of dialkenylhydroxylamines through *N*-vinylfluorenyl nitrones with monosubstituted allenoates.[15] The key intermediate for these strategies was benzoazepine, which has illustrated appealing versatility for the synthesis of indole derivatives by further transformations. In 2020, we developed a commercial gimeracil-catalyzed formal [4 + 1] cycloaddition of *N*-vinylfluorenyl nitrones with allenoates to prepare a series of spirofluorenylpyrrolines in good yields, showing that allenoates serve as a one-carbon synthon for the ring formation.[16] During the study of the [3 + 2] cycloaddition of *α,β*-unsaturated nitrones with various dipolarphiles in our group,[17] we surmised that metal catalysts, such as the inexpensive nickel(II) catalyst, might control a new transformation of benzoazepine intermediates owing to the easy coordination to the dicarbonyl group of benzoazepine. Herein we report that *α,β*-unsaturated nitrones reacted with allenoates, affording *N*-vinylindoles and *N*-vinylpyrroles in the presence of a nickel(II) catalyst under mild reaction conditions through a cascade of [3 + 2] cycloaddition, [3,3]-rearrangement, retro-Mannich reaction, condensation, and 1,4-addition in a one-pot reaction (Scheme 1C).

Initially, nitrone **1a** and allenoate **2a** were chosen as model substrates to explore the optimal reaction conditions. As shown in Table 1, nitrone **1a** reacted with allenoate **2a** in toluene at room temperature for 24 h, affording the desired *N*-vinylindole **3aa** in 26% yield with 1:1 Z/E isomers for the enamine bond (Table 1, entry 1). Solvent screening showed that dichloromethane (DCM) gave a higher yield than tetrahydrofuran (THF) and dichloroethane (DCE), and **3aa** was not observed in acetonitrile (MeCN) or dimethyl sulfoxide (DMSO) (Table 1, entries 2–6). Adding Lewis acid catalysts, such as $FeCl_3$ and $ZnCl_2$, delivered **3aa** in 28 and 40% yield, respectively (Table 1, entries 7 and 8). These results showed that adding a Lewis acid catalyst completely improved the Z/E ratio of **3aa**. Pleasingly, $NiCl_2$ as a catalyst improved the yield of **3aa** to 65% with a Z isomer only, which was determined by its 2D nuclear Overhauser effect spectroscopy (NOESY) spectra (Table 1, entry 9). The influence of N-ligands was also evaluated (Table 1, entries 10–13). The addition of 2,2-bipyridine (**L1**) and 1,10-phenanthroline (**L2**) obviously improved product **3aa** to 81 and 74% yield, respectively, whereas 2,9-dimethyl-1,10-phenanthroline (**L3**) and Pybox (**L4**) did not improve the yield, but the Z/E ratio of the enamine bond remained as 20:1. These results showed a higher reactivity of **L1** and **L2** than those using **L3** and **L4**, perhaps owing to the steric effect. The addition of $MgSO_4$ as an additive to the standard conditions afforded **3aa** in 83% yield (Table 1, entry 14). However, adding NaOH completely prohibited the formation of **3aa** (Table 1, entry 15). Therefore, the optimal conditions for preparing **3aa** were $NiCl_2$ (20 mol %), **L1** (24 mol %) as a ligand, and $MgSO_4$ as an additive in DCM at room temperature (rt).

With the optimized conditions in hand, we then explored the reaction scope with a variety of *α,β*-unsaturated nitrones **1** (Scheme 2). Various dibenzylideneacetone-derived nitrones **1a**–**1h** with electron-donating and electron-withdrawing groups at the para, meta, and ortho positions of the aryl ring smoothly afforded the corresponding products **3aa**–**3ha** in good yields. When 1.0 g of nitrone **1a** (3 mmol) was carried out under the standard conditions, **3aa** was obtained in 67% yield. Nitrone **1i** tolerated a 2-thienyl heteroaryl group, delivering **3ia** in 59% yield. Chalcone-derived nitrones **1j**

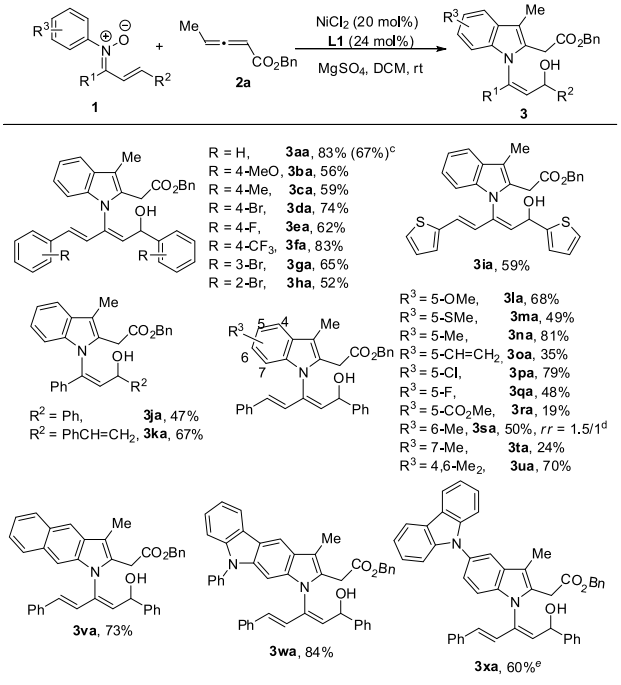
Table 1. Optimization of the Reaction Conditions[a]



entry	cat/**L**	solvent	**3aa** (Z/E)[b], yield (%)[c]
1		toluene	26 (1:1)
2		THF	32 (2:1)
3		DCE	42 (2:1)
4		DCM	58 (2:1)
5		MeCN	<5
6		DMSO	<5
7	$FeCl_3$	DCM	28 (10:1)
8	$ZnCl_2$	DCM	40 (10:1)
9	$NiCl_2$	DCM	65 (20:1)
10	$NiCl_2$/**L1**	DCM	81 (20:1)
11	$NiCl_2$/**L2**	DCM	74(20:1)
12	$NiCl_2$/**L3**	DCM	51 (20:1)
13	$NiCl_2$/**L4**	DCM	56 (20:1)
14[d]	$NiCl_2$/**L1**	DCM	83 (20:1)
15[e]	$NiCl_2$/**L1**	DCM	<5

[a]Reaction conditions: **1a** (0.2 mmol), **2a** (0.4 mmol), cat. (20 mol %), **L** (24 mol %), solvent (2.0 mL), rt, 16–24 h. [b]Ratio of Z and E for the enamine bond in **3aa**. [c]Isolated yield. [d]Adding $MgSO_4$ (200 mg). [e]Adding NaOH (2.0 equiv).

Scheme 2. Substrate Scope of *N*-Aryl Nitrones 1[a,b]



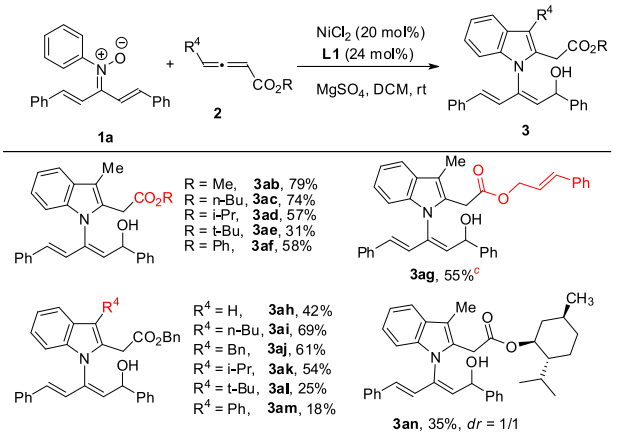
[a]Reaction conditions: **1** (0.2 mmol), **2a** (0.4 mmol), $NiCl_2$ (20 mol %), **L1** (24 mol %), $MgSO_4$ (200 mg), DCM (2.0 mL), rt, 16–72 h. [b]Isolated yield. [c]**1a** (1.0 g, 3.0 mmol). [d]rr = rearrangement regioselectivity. [e]Ran at 60 °C.

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with a phenyl and **1k** with a styrenyl at the R^2 group both reacted with **2a**, smoothly affording **3ja** and **3ka** in 47 and 67% yield, respectively. In addition to α,β-unsaturated ketone moieties, *N*-aryl groups were also evaluated. The substituents of *N*-aryl moieties could be present with electron-donating and electron-withdrawing groups at the para, meta, and ortho positions of the aryl ring, such as thioether, vinyl, halide, and ester groups, giving the corresponding products **3la**–**3ua** in moderate to good yields. When the R^3 of nitrone **1s** was a methyl group at the meta position, the [3,3]-rearrangement regioselectivity was 1.5/1. Pleasingly, nitrone **1v** with 2-naphthalenyl furnished **3va** in 73% yield as a single isomer with high [3,3]-rearrangement regioselectivity. When carbazole-derived nitrones **1w** and **1x** were carried out under the standard conditions, products **3wa** and **3xa** were obtained in 84 and 63% yield, respectively.

Next, the scope of allenoates was also studied. As shown in Scheme 3, a variety of allenoates **2** smoothly reacted with

Scheme 3. Substrate Scope of Allenoates 1[a,b]

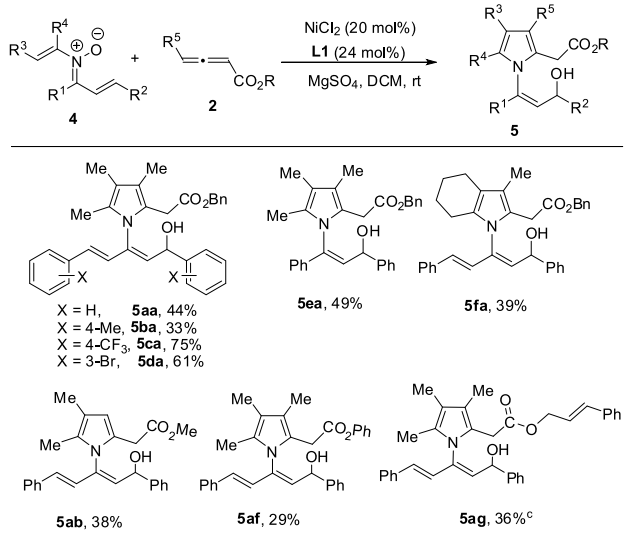


[a]Reaction conditions: **1a** (0.2 mmol), **2** (0.4 mmol), $NiCl_2$ (20 mol %), **L1** (24 mol %), $MgSO_4$ (200 mg), DCM (2.0 mL), rt, 16–72 h. [b]Isolated yield. [c]Ran at 60 °C.

nitrone **1a** to afford the corresponding products **3ab**–**3an** in moderate to good yields ranging from 18 to 79%. The R group of allenoate could be present with alkyl, aryl, and alkenyl groups. When the R group was *i*-Pr or *t*-Bu, the yields of **3ad** and **3ae** dropped to 57 and 31% yield, respectively. These results indicated that the steric effect of the R group had a great impact on the yield of *N*-vinyl indoles. The R^4 group of allenoate was also evaluated. Interestingly, when unsubstituted allenoate **2h** was carried out under the standard conditions, *N*-vinylindole **3ah** was also obtained in 42% yield, and the corresponding dihydrocarbazole and dihydropyridoindole scaffolds reported in Anderson's method were not observed in this case. The *i*-Pr-, *t*-Bu-, and Ph-substituted allenoates (**2k**–**2m**) all gave lower yields (**3ak**–**3am**). These experimental results demonstrated that the steric hindrance of R^4 had a significant effect on the formation of *N*-vinylindoles. When a (−)-menthol derived allenoate **2n** was subjected to the standard conditions, product **3an** was obtained in 35% yield with a 1:1 diastereoselectivity.

In addition to *N*-aryl nitrones, *N*-vinyl nitrones were also tested with **2a** to determine the reaction compatibility. As shown in Scheme 4, it was found that various *N*-vinyl-α,β-unsaturated nitrones **4a**–**4f** and allenoates (**2a**, **2b**, **2f**, **2g**)

Scheme 4. Scope of *N*-Vinyl Nitrones 4[a,b]

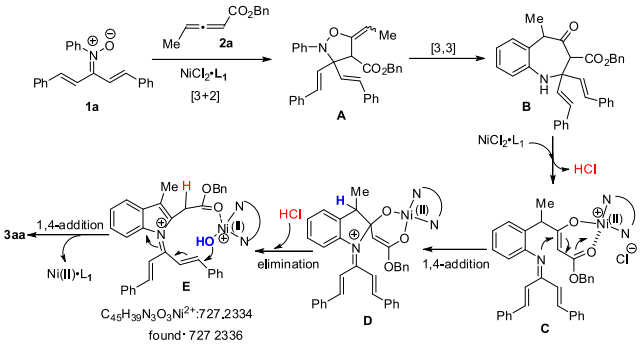


[a]Reaction conditions: **4** (0.2 mmol), **2** (0.4 mmol), $NiCl_2$ (20 mol %), **L1** (24 mol %), $MgSO_4$ (200 mg), DCM (2.0 mL), rt, 16–24 h. [b]Isolated yield. [c]Ran at 60 °C.

proceeded smoothly to afford the corresponding polysubstituted *N*-vinylpyrroles **5aa**–**5ag** in moderate yields. In these cases, the yields of *N*-vinylpyrroles were lower than those of *N*-vinylindoles because of the easy decomposition of *N*-vinyl nitrones to the corresponding ketones in the presence of a nickel(II) catalyst. This method provides an alternative approach to the preparation of polysubstituted *N*-vinylpyrroles.

On the basis of the O^{18}-labeled experimental results suggesting that the hydroxyl group in **3aa** might be derived from water and the detection of intermediates by high-resolution mass spectrometry (HRMS) trace experiments,[18] a possible mechanism for the formation of *N*-vinylindole **3aa** from **1a** and **2a** was proposed (Scheme 5). Initially, the $NiCl_2$·

Scheme 5. Proposed Mechanism



L1 complex was immediately formed when $NiCl_2$ was mixed with **L1**.[19] Then, nitrone **1a** was activated by the $NiCl_2$·**L1** catalyst and underwent a [3 + 2] cycloaddition with allenoate **2a** to form intermediate **A**. A [3,3]-rearrangement of **A** afforded benzoazepine **B**, which underwent a retro-Mannich reaction and enolation of the 1,3-dicarbonyl group in the presence of $NiCl_2$·**L1** to give imine **C** and HCl. Imine **C** underwent an intramolecular aza-Michael addition activated by Ni(II) to form **D**. Intermediate **D** went through an elimination and was trapped by HCl to form indole scaffold **E**, which was

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supported by HRMS. (See the O^{18}-labeled experiment in the Supporting Information.) Intermediate **E** facilitated an intramolecular 1,4-addition from one side of the styrenyl to give *N*-vinylindole **3aa** with high Z-selectivity of the enamine bond owing to the carbonyl group serving as a directing group and finally released the Ni(II)·**L1** catalyst to finish the catalytic cycle.

In summary, we have identified a nickel(II)-catalyzed [3 + 2] cycloaddition of *α,β*-unsaturated nitrones with allenoates to prepare a variety of *N*-vinylindoles and *N*-vinylpyrroles in moderate to good yields under mild reaction conditions. The reaction tolerated diverse nitrones and allenoates with various functional groups. Mechanistic studies revealed that the nickel(II) catalyst played an important role in the reaction selectivity. This method features mild reaction conditions, a broad substrate scope, a highly atom-economical reaction, and high Z-selectivity of the enamine bond.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c02695.

Detailed mechanistic studies, experimental procedures, characteristic data, and copies of NMR spectra of compounds **3** and **5** (PDF)

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Notes

The authors declare no competing financial interest.

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(18) See more details of the mechanistic studies and the intermediate detection by the HRMS trace experiments for the reaction of nitrone **1a** with allenoate **2a** in the Supporting Information.

(19) See more details of the preparation and reactivity of the $NiCl_2$·**L1** complex in the Supporting Information. For the synthetic procedure of the $NiCl_2$·**L1** complex, also see: Singh, K.; Kabadwal, L. M.; Bera, S.; Alanthadka, A.; Banerjee, D. Nickel-Catalyzed Synthesis of N-Substituted Pyrroles Using Diols with Aryl- and Alkylamines. *J. Org. Chem.* **2018**, *83*, 15406.

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